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Pogo Pin Inspection Automation • Capston Project

Cohu | UC San Diego JACOBS SCHOOL OF ENGINEERING Department of Mechanical and Aerospace Engineering

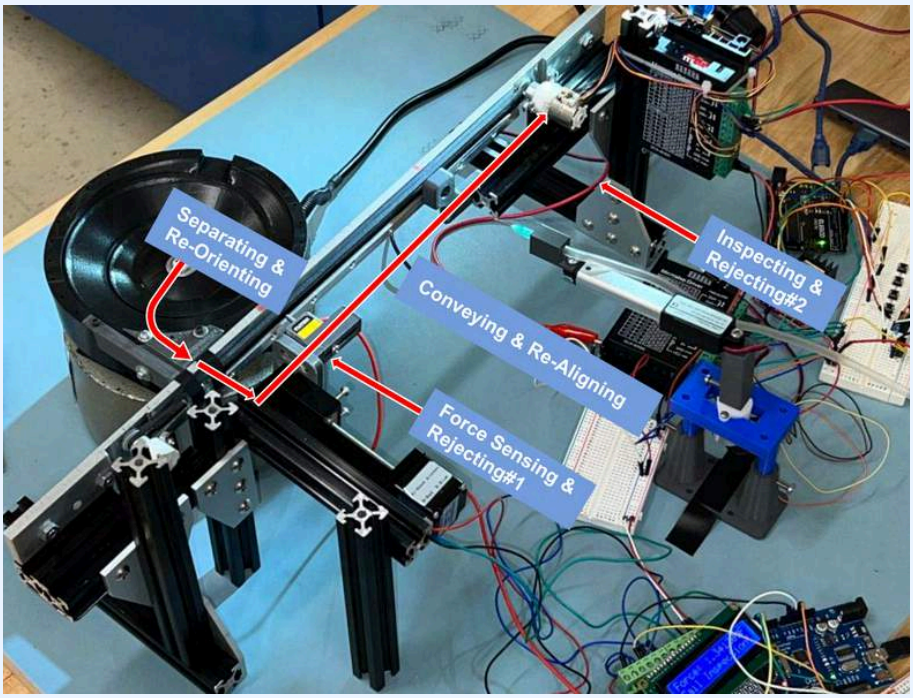
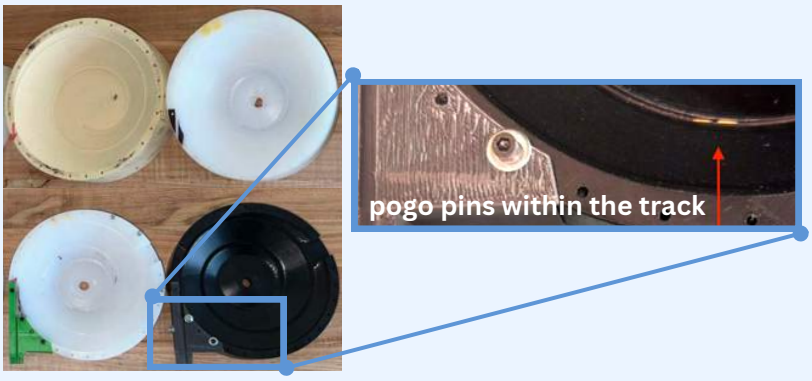


Figure 1, assembled system with labeled operation process.

Figure 2, magnified photo of the length 3mm, OD 0.3 mm pogo pin.

Project Website

Figure 3, my team and our advisor at Capstone Conference



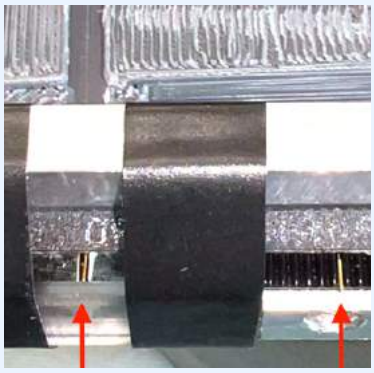


Figure 4, 5, 6, 7 (Left to right), design iterations, subsystem hands off design, assembled bowl feeder system

A four-step inspection system that automated inspection process for pogo pins.

- Designed a vibratory bowl feeder system:
  - adapted rapid design iteration and achieved 5 design iterations in 11 weeks.
  - capable of sorting 5000+ part continuously.
  - in-house produced the system and resulted in over $15,000 reduction in expense.
- Architected the overall system design and managed system integration.
- Led internal design meetings and external reviews with sponsor and advisors.
- Won 1st place in Capstone Project Conference, recognized by ASME and alumni judges.

Skills:

Design | Modeling | Prototyping | Testing | Solidworks | Leadership | Stage-Gate Process | BOM

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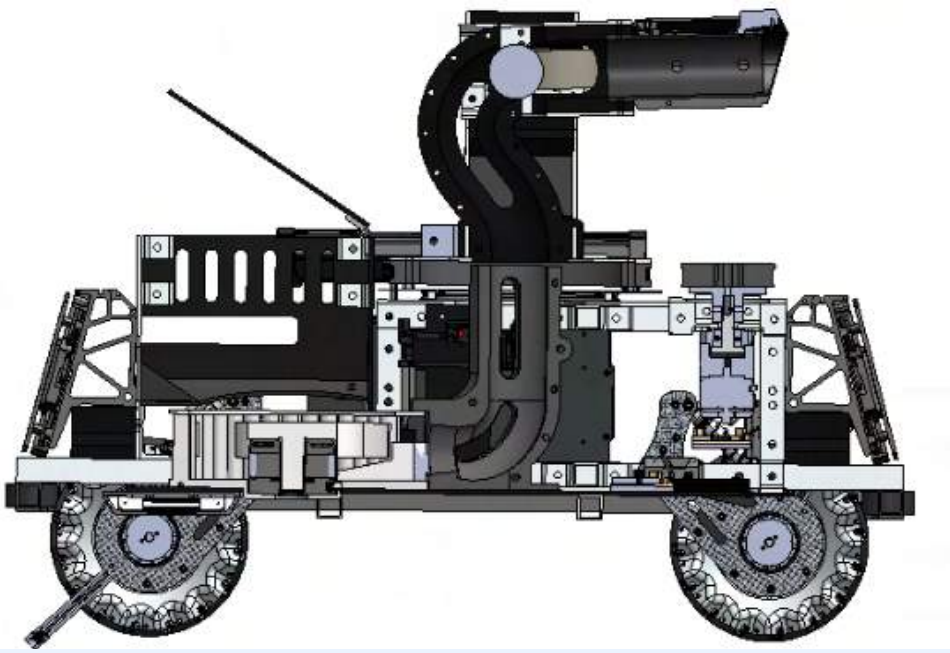
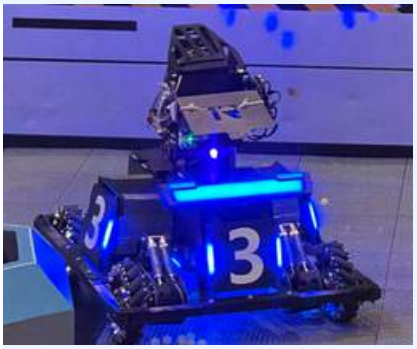
Rotary Indexer System

- Hero Robot

TRITION ROBOTICS





Link to the project website

Figure 1, 2, side view of the robot and robot in action

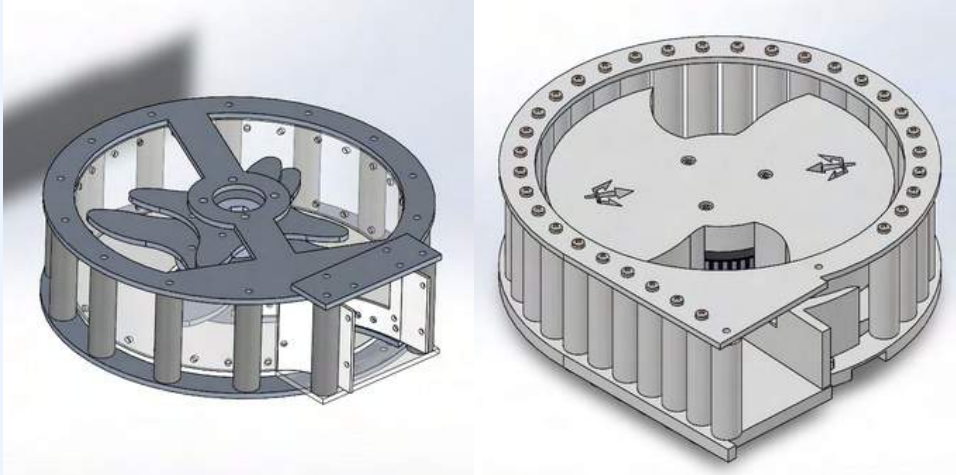


Figure 3, design evolution

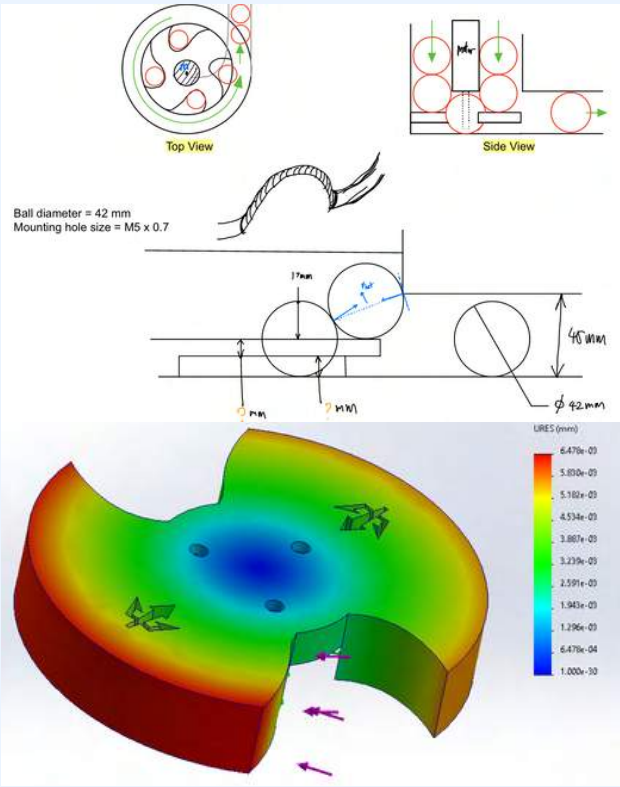


Figure 4, 5, Force analysis and Finite Element Analysis Simulations (Solidworks)

A rotary indexer system that supplies projectiles for battle robot's turret.

- Led the full production cycle from brainstorming idea to fabrication.
- Collaborated with electrical engineers to ensure safety limits of electrical components.
- Conducted force and FEA analysis, ensured structural stability of the design.
- Optimized the projectile feeding paths, reduced friction at bending corners.
- Assisted the team ranking 2nd place in the RoboMaster University League 2022.

Skills:
Design for manufacturing | GD&T | Solidworks FEA | Power & Force analysis | 3D-printing

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ASL Reading Glove

- HARD Hack Challenge 2023

IEEE | HKN UC San Diego JACOBS SCHOOL OF ENGINEERING

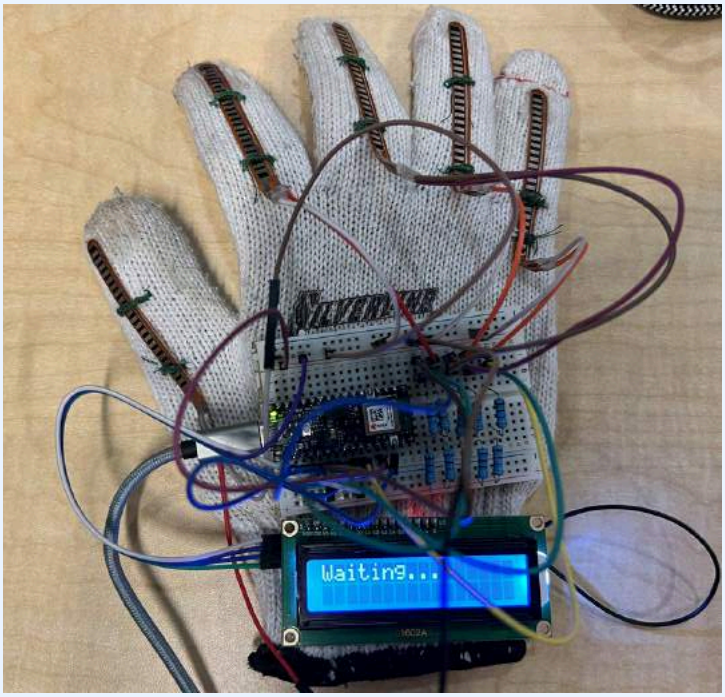

Figure 1, top view of the glove

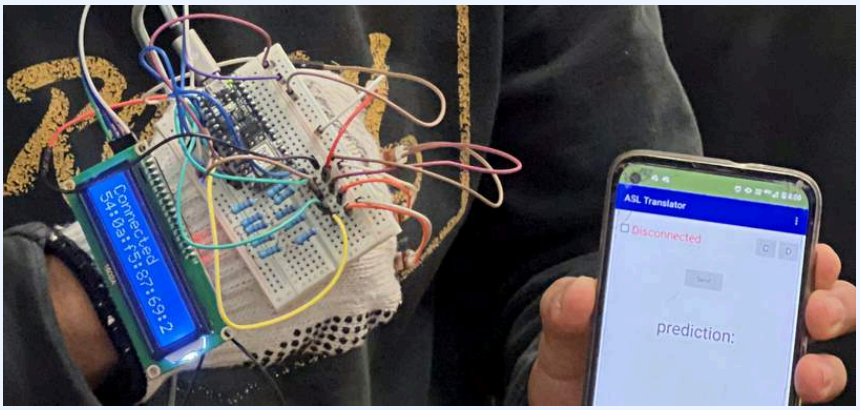

Figure 2, demostration of the glove in use

Figure 3, soldered flex sensor

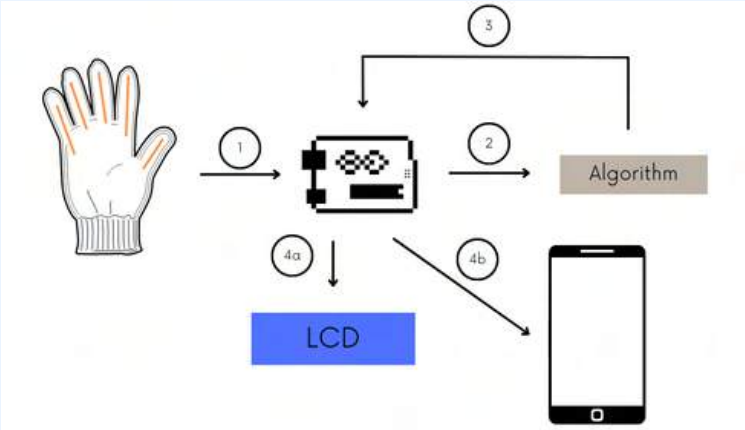

Figure 4, design schematic

A glove that transcribes American Sign Language (ASL) alphabets and display on an LCD screen and remotely on a phone application.

- Designed the mechanical mounting of flex sensors, ensured feasibility of data collection.
- Mapped recorded data from each sensor to reflect finger joint motion angle.
- Adapted fast pace design environment and achieved complete product in 24 hours.
- Won 1st place in the hardware hackathon, recognized by IEEE and alumni judges.

Skills:
Design for Manufacturing | Arduino Nano BLE | Flex Sensors | Interdisciplinary Collaboration

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Transmission Circular Polariscopy System • Mechanics of Soft Materials Lab

UCLA Samueli School of Engineering
SUMMER UNDERGRADUATE RESEARCH PROGRAM

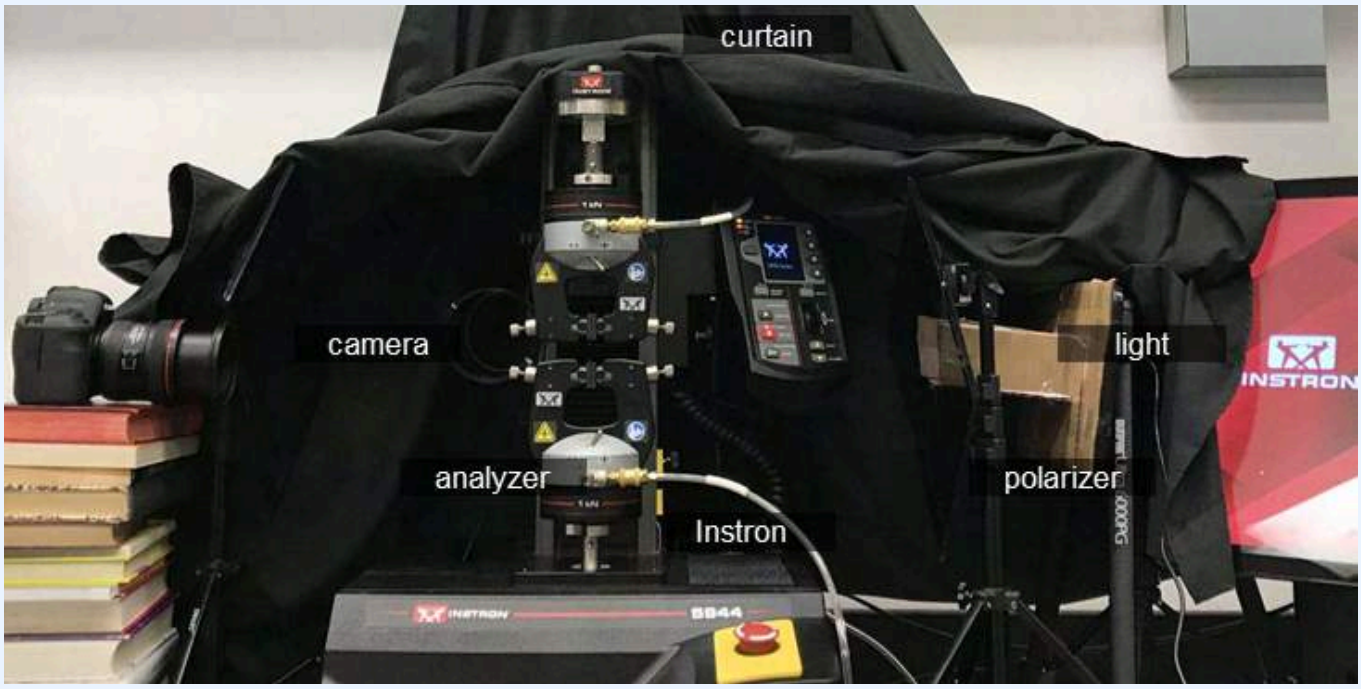

Figure 1, picture of the TCP testing system.

Published Paper: Rate-dependent stress-order coupling in main-chain liquid crystal elastomers

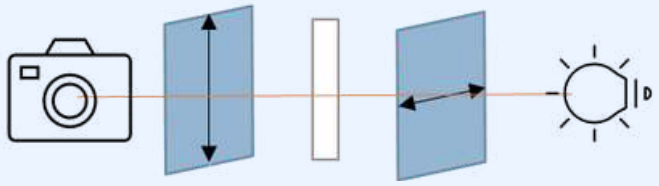

Figure 2, skematics of the TCP system.

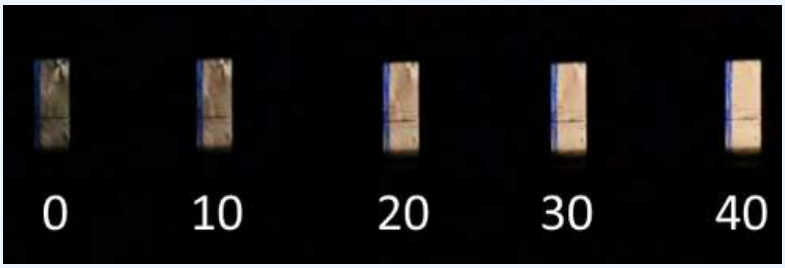

Figure 3, behavior at different polarization angle.

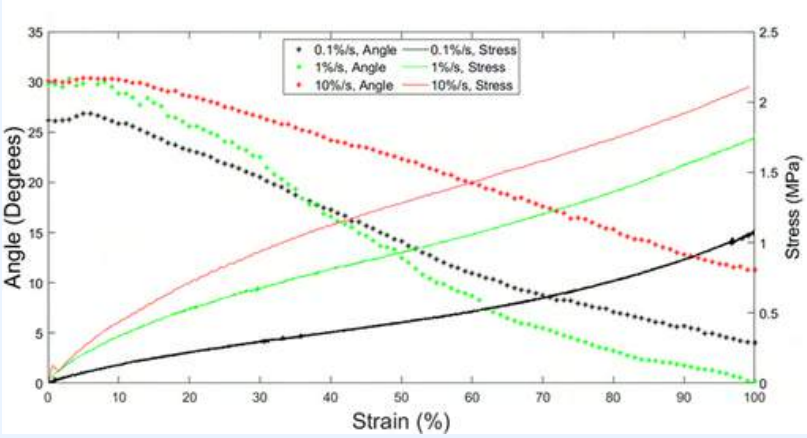

Figure 4, director angles' reaction during stress-strain tests at various initial angles, overlaying with the stress-strain relationship.

A polarization system with variable light transmissivity adjustment to observe material transformation during mechanical load tests.

- Performed stress-strain and strain rate testing using Instron machine.
- Analyzed material using stress-strain and strain-rate testing.
- Automated data analysis using ImageJ & Matlab, accelerated research progress 3 weeks.

Skills:
Mechanical Design | Tensile Test | MatLab | ImageJ | Automation | Graph Analysis

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Pick-Lift Robot

- MAE3: Engineering Design & Graphics

UC San Diego
JACOBS SCHOOL OF ENGINEERING
Department of Mechanical and Aerospace Engineering

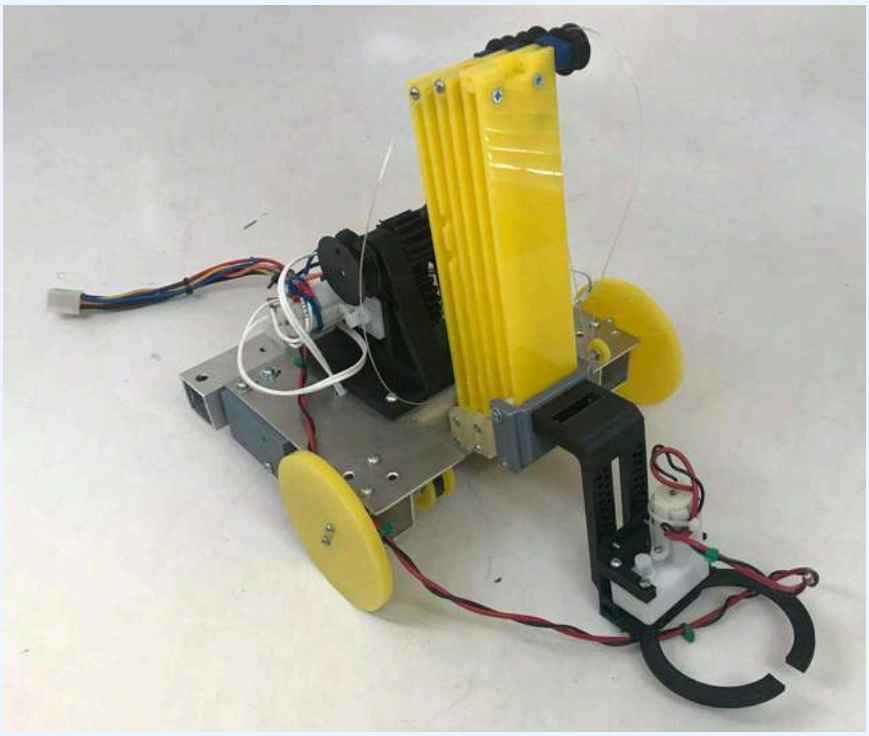
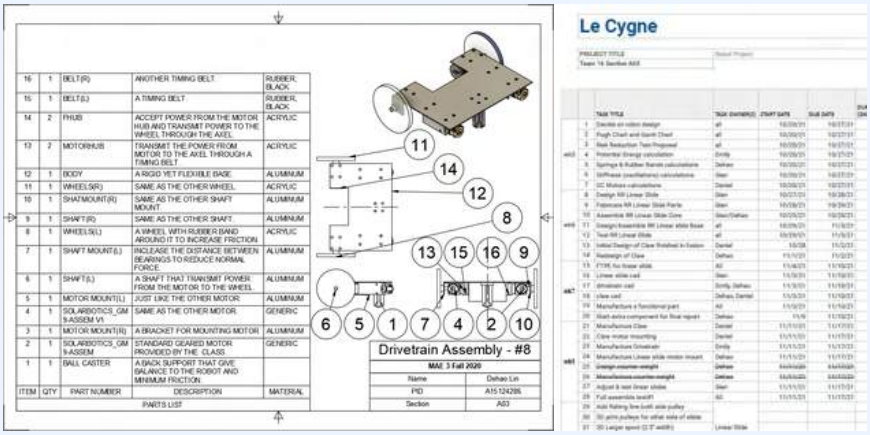


Figure 1, picture of the finalized robot

Figure 2, spool design

Link to the project website



ITEM	QTY	PART NUMBER	DESCRIPTION	MATERIAL
16	1	BELT(R)	ANOTHER TIMING BELT	RUBBER, BLACK
15	1	BELT(L)	A TIMING BELT	RUBBER, BLACK
14	2	FHUB	ACCEPT POWER FROM THE MOTOR HUB AND TRANSMIT POWER TO THE WHEEL THROUGH THE AXEL	ACRYLIC
13	2	MOTORHUB	TRANSMIT THE POWER FROM MOTOR TO THE AXEL THROUGH A TIMING BELT	ACRYLIC
12	1	BODY	A RIGID YET FLEXIBLE BASE	ALUMINUM
11	1	WHEEL(R)	SAME AS THE OTHER WHEEL	ACRYLIC
10	1	SHAFMOUNT(R)	SAME AS THE OTHER SHAFT MOUNT	ALUMINUM
9	1	SHAFT(R)	SAME AS THE OTHER SHAFT	ALUMINUM
8	1	WHEEL(L)	A WHEEL WITH RUBBER BAND AROUND IT TO INCREASE FRICTION	ACRYLIC
7	1	SHAFT MOUNT(L)	INCLEASE THE DISTANCE BETWEEN BEARINGS TO REDUCE NORMAL FORCE	ALUMINUM
6	1	SHAFT(L)	A SHAFT THAT TRANSMIT POWER FROM THE MOTOR TO THE WHEEL	ALUMINUM
5	1	MOTOR MOUNT(L)	JUST LIKE THE OTHER MOTOR	ALUMINUM
4	1	SOLARBOTICS_GM 9-ASSEM V1	SAME AS THE OTHER MOTOR	GENERIC
3	1	MOTOR MOUNT(R)	A BRACKET FOR MOUNTING MOTOR	ALUMINUM
2	1	SOLARBOTICS_GM 9-ASSEM	STANDARD GEARED MOTOR PROVIDED BY THE CLASS	GENERIC
1	1	BALL CASTER	A BACK SUPPORT THAT GIVE BALANCE TO THE ROBOT AND MINIMUM FRICTION	

PARTS LIST

Figure 4&5, drivetrain draft paper, BOM, and Gantt Chart

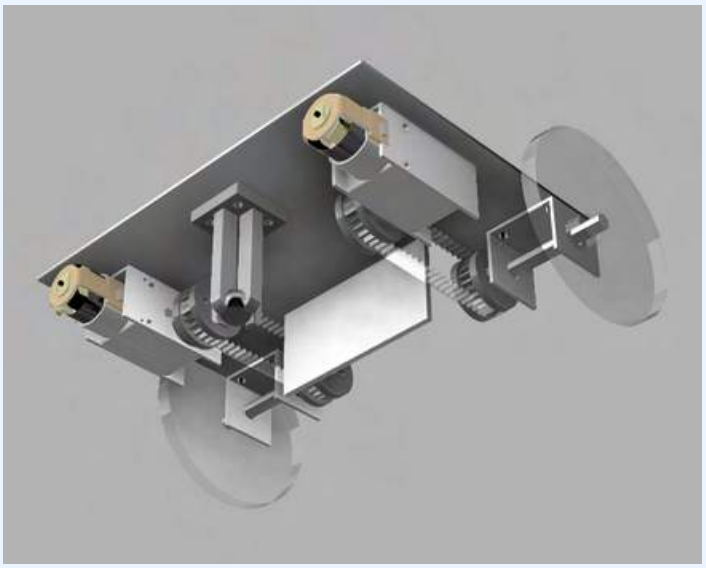


Figure 3, drivetrain design

A pick-lift robot with maneuver capability over obstacle surrounded terrain.

- Performed risk analysis on critical components, prevented 2 major design flaws.
- Performed power & gear analysis on DC motors, designed an agile drivetrain.
- Led design & deliverables for the team of 4, achieved on-time project completion.
- Won the best manufacturer award out of 41 teams.

Skills:
Fusion 360 | Laser Cutting | Fabrication | AutoCAD | 3D Printing | GD&T | Power/Force Analysis

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Trocar Claw

- Engineer Week Design Challenge

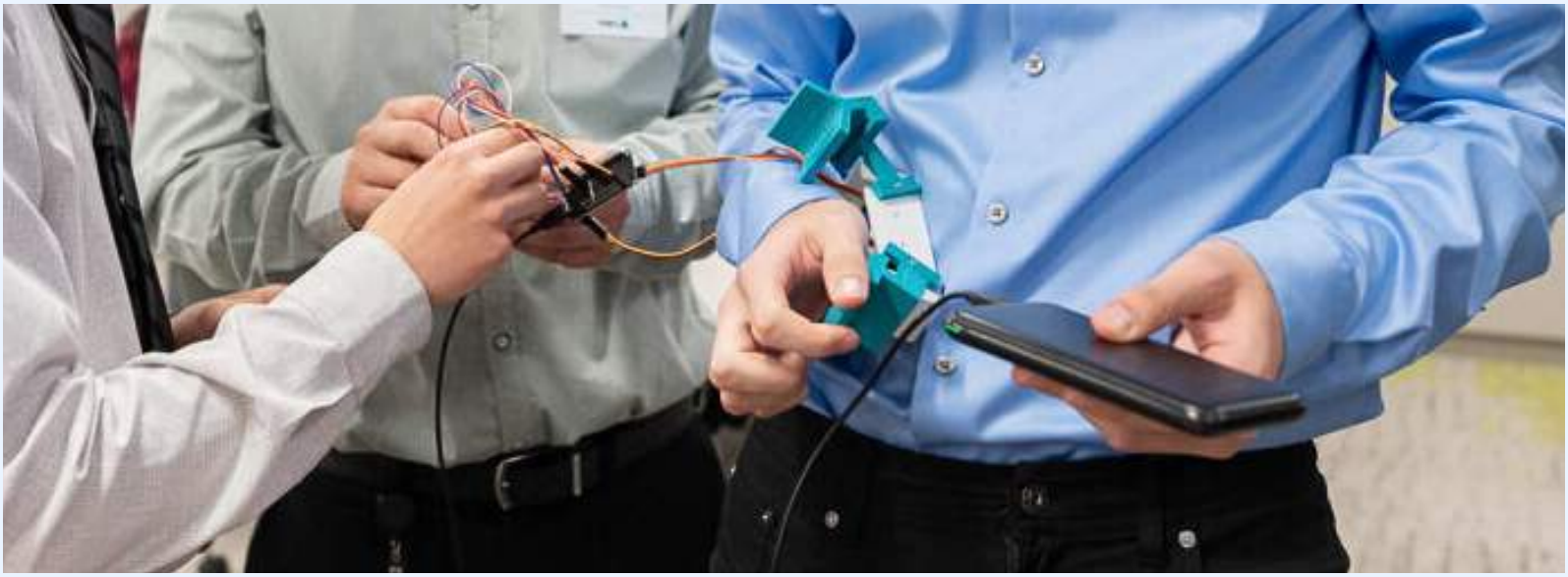

Figure 1, Trocar gripper in action

Figure 2, Design presentation

Figure 3, Prototype testing

A 3D-printed surgery trocar gripper prototype for surgical operations.

- Designed the mechatronic system using microcontroller and potentiometers, achieved the operation requirement of gripper using minimum materials.
- Performed risk analysis to assess optimal gear output, achieved sufficient friction force using materials with low coefficient of friction.
- Led a group of 4 engineers, collaboratively created design planning and correction.

Skills:
ESP32 | Servo Motor | Linear Circuit | Potentiometer | Force-balance analysis | Risk Analysis

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Machining Projects

UCSD MAE Machine Shop UC San Diego JACOBS SCHOOL OF ENGINEERING Department of Mechanical and Aerospace Engineering

Figure 1, Geneva Drive

A Geneva drive made from HDPE plastic and Aluminum 6061.

- Wrote CNC tool path with CAM software.
- Practiced tolerance setting, manual machining, and component assembling.
- Learned machining safety practice.
- Gain practical experience of manufacture for design.

Skills:
Fusion 360 CAM | Tormach 440 Mill | Manual Mill & Lathe

Figure 2, Testing with air blow gun

The bowl unit for vibratory bowl feeder, made from UHMW plastic.

- Designed CAD models for manufacturability.
- Wrote CNC tool path for a complex geometry, completed a 4-axis job in a 3 axis NC machine.
- Explored machine setting for different materials and selected optimal tools.
- Practiced tolerance setting and quality control.
- Frequently consult with machining experts to improve machining process.

Skills:
Fusion 360 CAM | HAAS Mini Mill 2 | Manual Mill & Lathe

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Innovation for National Security

- MCAS - Miramar

Figure 1, end of project team picture

Official LinkedIn Post

Identified the source problems that impacted Marines' quality of life and mission readiness at Marine Corps Air Station, Miramar (MCAS - Miramar).

- Managed operation planning and progress summary for a team of 8.
- Collaboratively reached out and cold-emailed Marine enlisted personnels and officers, successfully establishing relationships that resulted in securing 82 interviews.
- Conducted interview analysis and navigated responses to validate problem hypotheses.
- Filtered and narrowed down to source problems are solvable without adding burden to base expenditure.

Skills:
Interdisciplinary Collaboration | Survey Interview | Presentation | Lean Business Model